



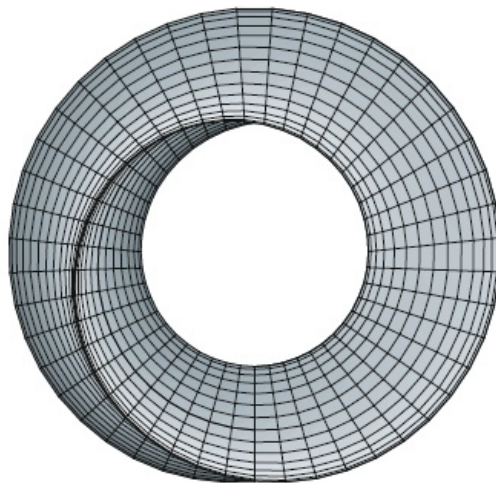
UNIVERSITY OF ZIMBABWE

HMTHCS111 CALCULUS 2

Calculus of Several Variables

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Abstract

Everybody is a genius. But if you judge a fish by its ability to climb a tree, it will live its whole life believing that it is stupid.

—Albert Einstein

These notes are not intended as a textbook. It is hoped however that they will minimize the amount of note-taking activity which occupies so much of a student's class time in most courses in Mathematics. Probably the most important aspect of the notes is the set of tutorials. You should develop the practice of attempting several of these problems every week. Many of the problems are quite difficult so please consult your tutor/lecturer if you are not blessed with success initially. Do not acquire the habit of abandoning a problem if it does not yield to your first attempt; a defeatist attitude is your greatest adversary. Solution of a problem, even with some assistance from the teacher when necessary, is a fine boost to your morale. You will find that a strong effort expended on the earlier part of the courses will be rewarded by growing self-confidence and easier success later.

The emphasis is on understanding concepts. I think that nearly everybody agrees that this should be the primary goal of calculus instruction. In fact, the impetus for the current calculus reform movement came from the Tulane Conference in 1986, which formulated as their first recommendation:

Focus on conceptual understanding.

I have tried to implement this goal through the *Rule of Three*: “Topics should be presented geometrically, numerically, and algebraically.” Visualization, numerical and graphical experimentation, and other approaches have changed how we teach conceptual reasoning in fundamental ways. The Rule of Three has been expanded to become the *Rule of Four* by emphasizing the verbal, or descriptive, point of view as well.

Chapter 1

What is Calculus?

A great discovery solves a great problem but there is a grain of discovery in the solution of any problem. Your problem may be modest; but if it challenges your curiosity and brings into play your inventive faculties, and if you solve it by your own means, you may experience the tension and enjoy the triumph of discovery.

—George Poyla

To a Roman in the days of the empire, a “calculus” was a pebble used in counting and gambling. Centuries later, “calulare” came to mean “to calculate,” “to compute,” “to figure out.” For our purposes, calculus is elementary mathematics (algebra, geometry, trigonometry) enhanced by the *limit process*.

It is fitting to say something about the history of calculus. The origins can be traced back to ancient Greece. The ancient Greeks raised many questions (often paradoxical) about tangents, motion, area, the infinitely small, the infinitely large, —questions that today are clarified and answered by calculus. Here and there the Greeks themselves provided answers (some very elegant), but mostly they provided only questions.

After the Greeks, progress was slow. Communication was limited, and each scholar was obliged to start almost from scratch. Over the centuries, some ingenious solutions to calculus-type problems were devised, but no general techniques were put forth. Progress was impeded by the lack of a convenient notation. Algebra, founded in the ninth century by Arab scholars, was not fully systematized until the sixteenth century. Then, in the seventeenth century, Descartes established analytic geometry, and the stage was set.

The actual invention of calculus is credited to Sir Isaac Newton (1642-1727), an Englishman, and to Gottfried Wilhelm Leibniz (1646-1716), a German. Newton’s invention is one of the few good turns that the great plague did mankind. The plague forced the closing of Cambridge University in 1665, and young Isaac Newton of Trinity College returned to his home in Lincolnshire for eighteen months

of meditation, out of which grew his method of fluxions, his theory of gravitation, and his theory of light. The method of fluxions is what concerns us here. A treatise with this title was written by Newton in 1672, but it remained unpublished until 1736, nine years after his death. The new method (calculus to us) was first announced in 1687, but in vague general terms without symbolism, formulas, or applications. Newton himself seemed reluctant to publish anything tangible about his new method, and it is not surprising that its development on the Continent, in spite of a late start, soon overtook Newton and went beyond him.

Leibniz started his work in 1673, eight years after Newton. In 1675 he initiated the basic modern notation: dx and $\int$. His first publications appeared in 1684 and 1686. These made little stir in Germany, but the two brothers Bernoulli of Basel (Switzerland) took up the ideas and added profusely to them. From 1690 onward, calculus grew rapidly and reached roughly its present state in about a hundred years. Certain theoretical subtleties were not fully resolved until the twentieth century.